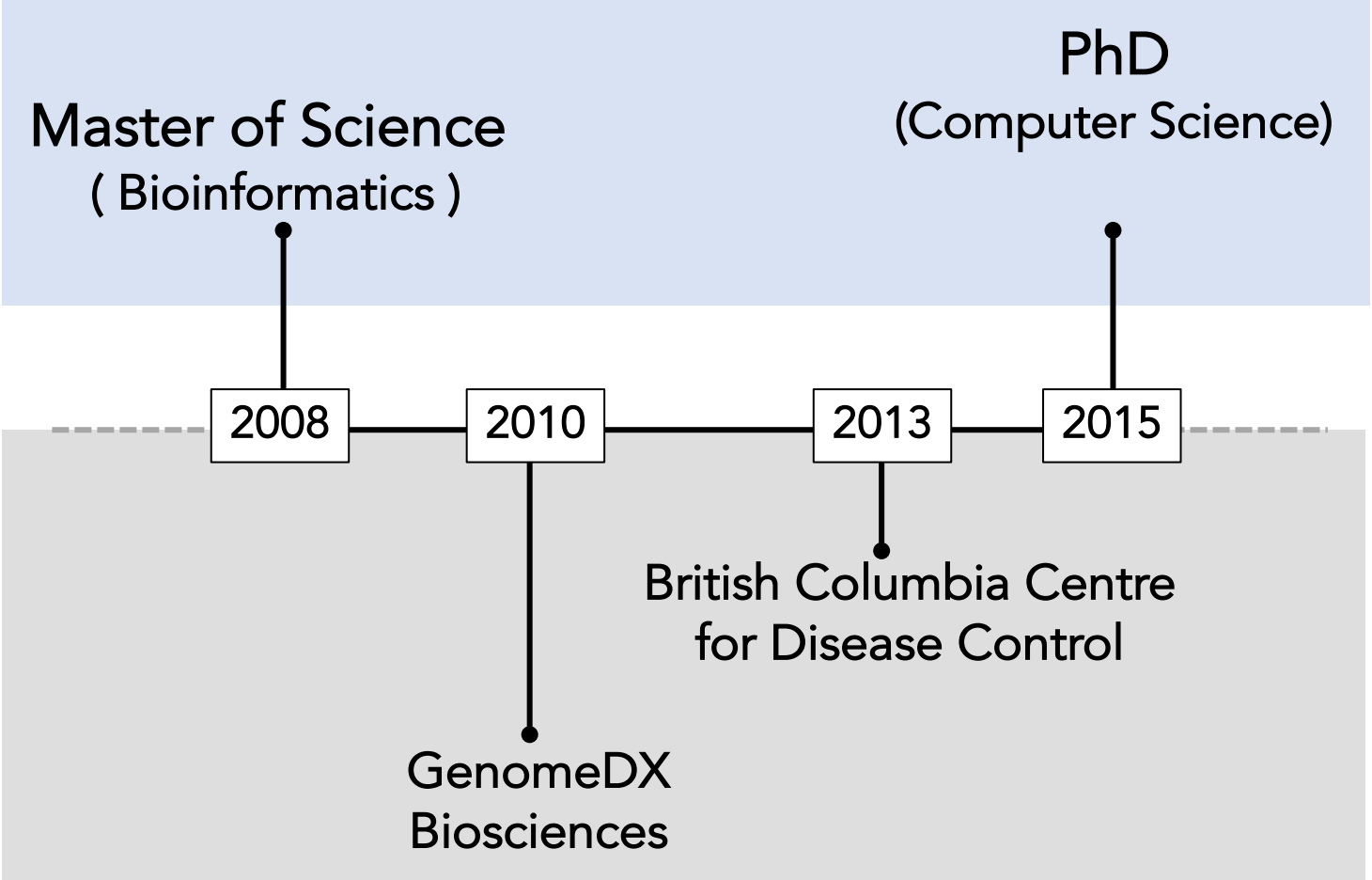
An Introduction to Data Visualization

Anamaria Crisan

@amcrisan acrisan@cs.ubc.ca http://cs.ubc.ca/~acrisan

PhD Candidate, Computer Science
University of British Columbia



Webinar Learning Goals

Yesterday

Have a high-level understanding of data visualization design and evaluation

Today

Have a basic understanding of different data visualizations tools as well as their strengths and limitations

A Brief Follow-up on some Questions from Yesterday

- **More on colour**
  - Designing color pallets with color brewer: http://colorbrewer2.org/
  - Thinking deeper about colour: http://www.cs.ubc.ca/~tmm/talks/minicourse14/vad17color-short.pdf
- **On Alberto Cairo and interpretation of visualizations**
  - See some of his talks on "Visual Trumpery" about misleading data visualizations

What we'll talk about

Data Visualization Tools

Tool #1 : Tableau

Tool #2 : Shiny (R)

A comment on *"How should we visualize data"*

There are two aspects of visualizations to think about:

How do you make a visualization?

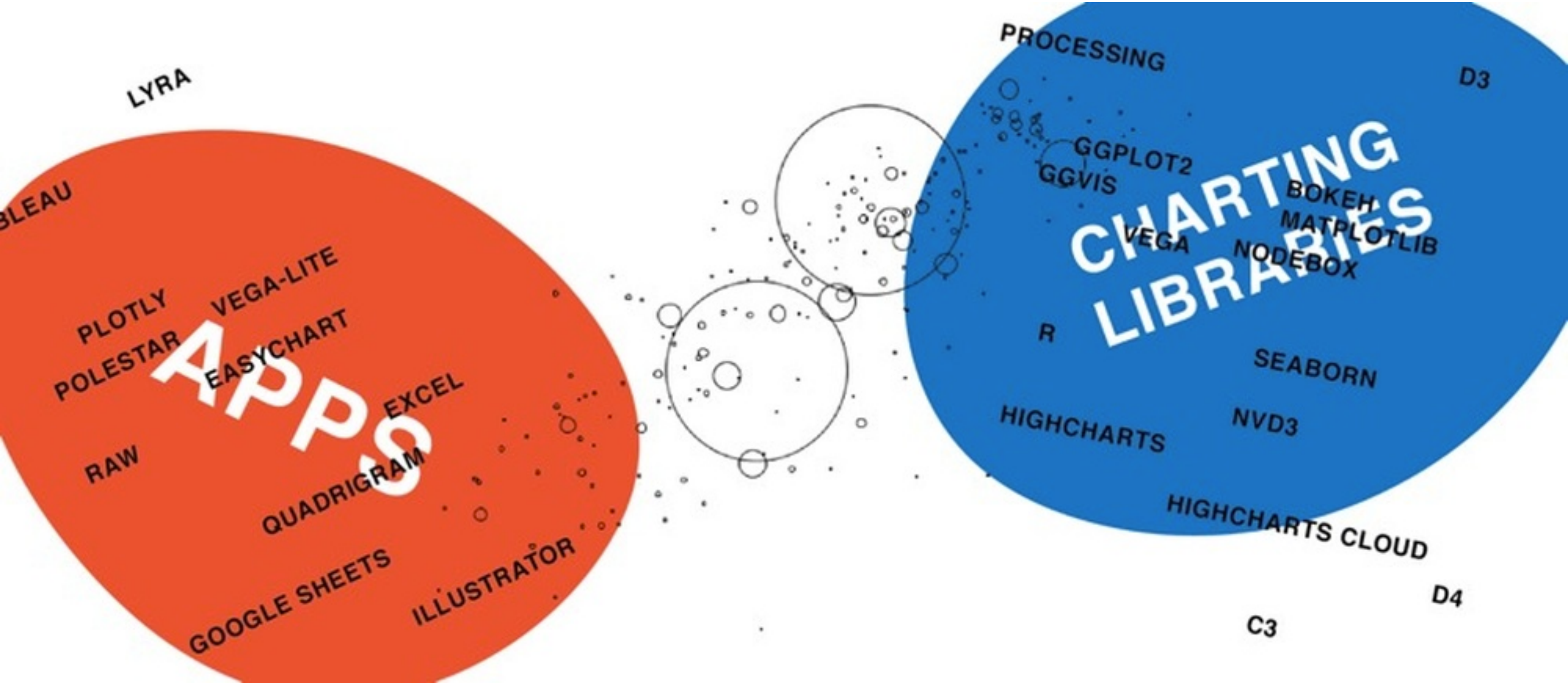
Is it the right visualization?

Data Visualization Tools

Tools & Libraries : An Overview

See this excellent post by Lisa Charlotte Rost : *http://bit.ly/2gRGx1J*

(figures taken from her post)

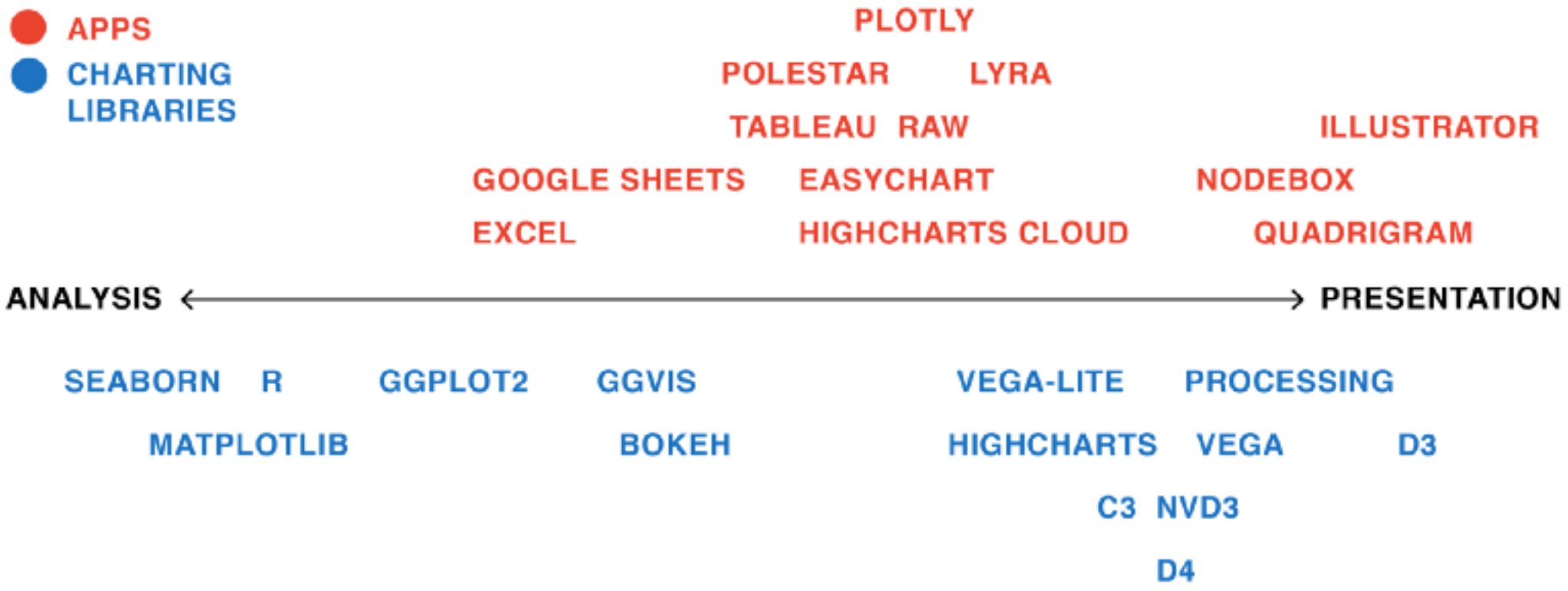


Tools & Libraries : An Overview

See this excellent post by Lisa Charlotte Rost : *http://bit.ly/2gRGx1J*

(figures taken from her post)

Analysis vs Presentation



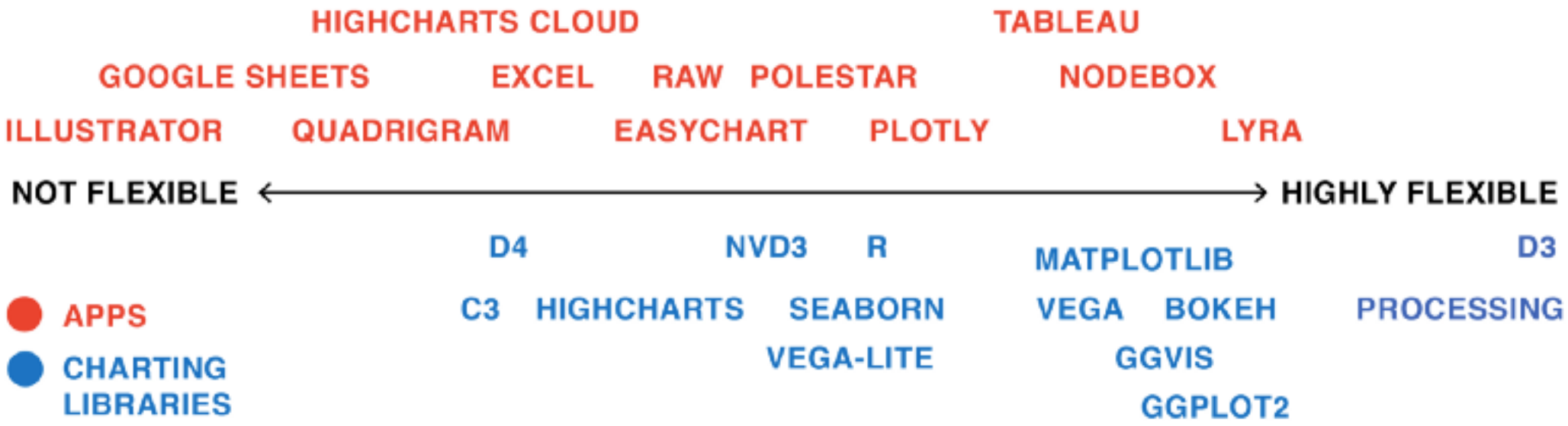
Tools & Libraries : An Overview

See this excellent post by Lisa Charlotte Rost : *http://bit.ly/2gRGx1J*

(figures taken from her post)

Extent of Flexibility

How easy/hard it is to make data visualizations (including custom/novel visualizations)



Tools & Libraries : An Overview

See this excellent post by Lisa Charlotte Rost : *http://bit.ly/2gRGx1J*

(figures taken from her post)

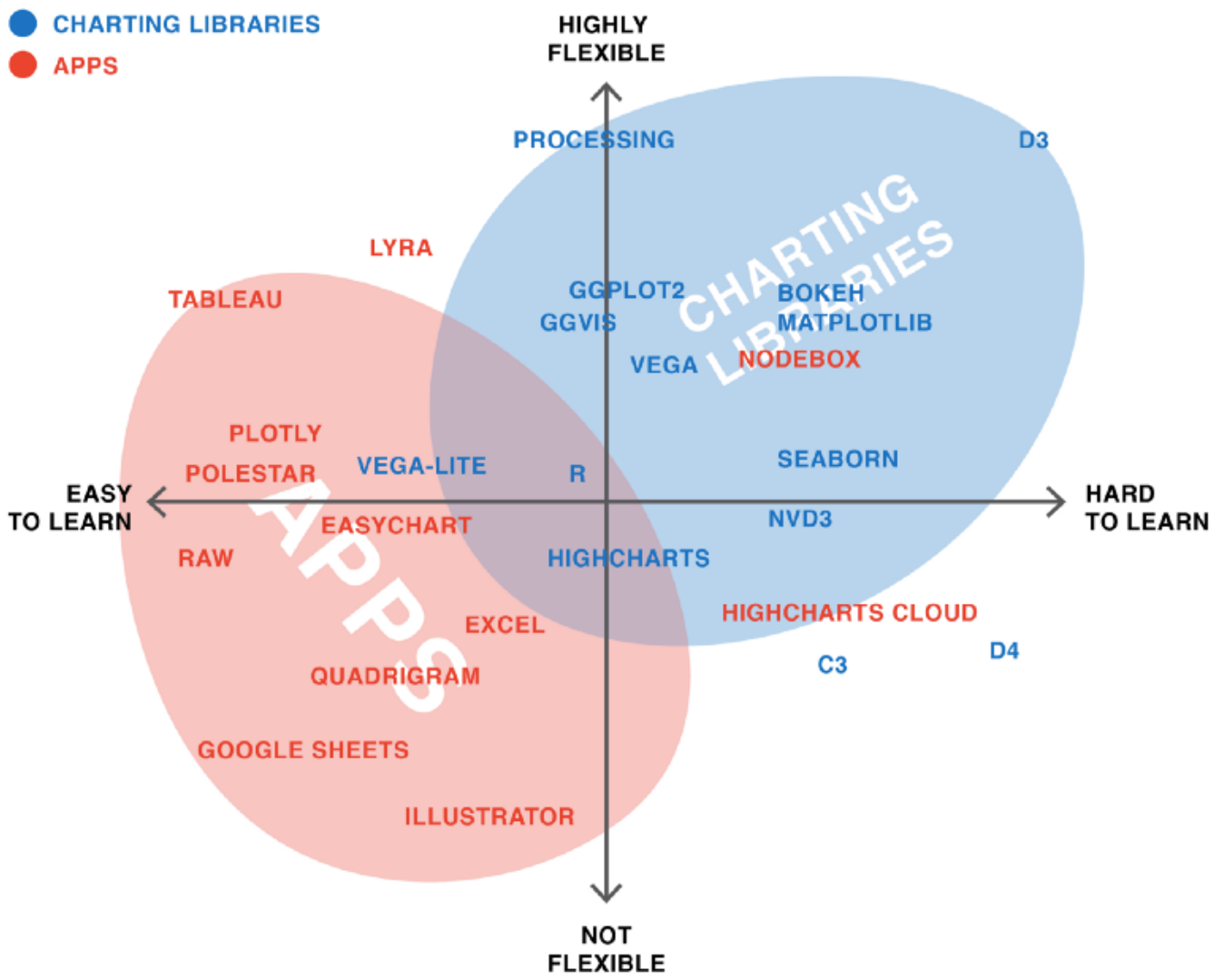
Static vs Interactive

	STATIC	WEB - INTERACTIVE
APPS	ILLUSTRATOR, NODEBOX, EXCEL, POLESTAR, RAW	HIGHCHARTS CLOUD, QUADRIGRAM, EASYCHRT, DATAWRAPPER, TABLEAU, PLOTLY, GOOGLE SHEETS
CHARTING LIBRARIES	GGPLOT2, MATPLOTLIB, R, SEABORN, BOKEH, PROCESSING	D3, D4, C3, NVD3, GGVIS, HIGHCHARTS, SHINY, VEGA, VEGA-LITE

Tools & Libraries : An Overview

See this excellent post by Lisa Charlotte Rost : *http://bit.ly/2gRGx1J*

(figures taken from her post)



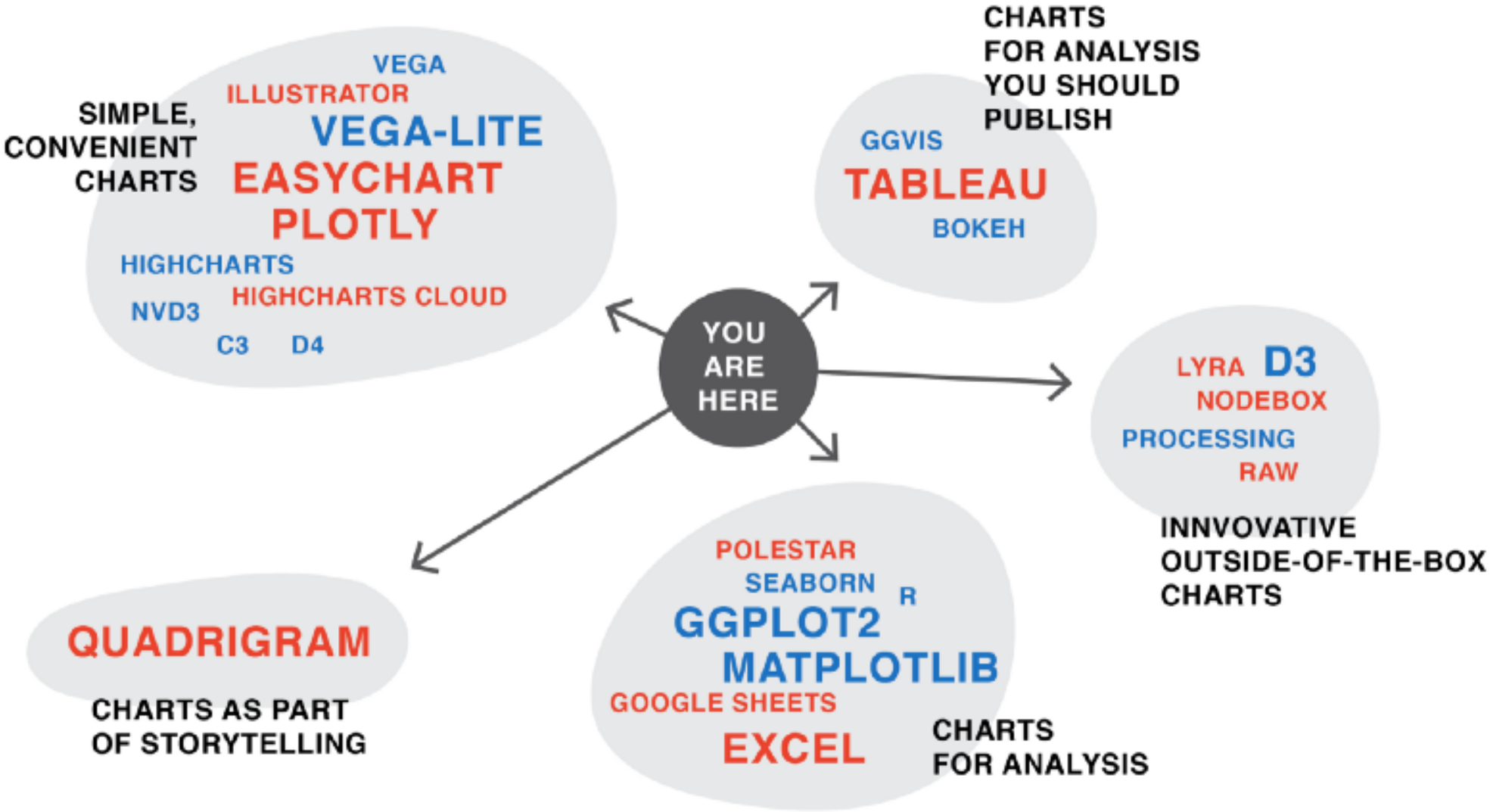
"There are no perfect tools, just good tools for people with certain goals"

See a detailed table here: http://bit.ly/2DeWPwV

Tools & Libraries : An Overview

See this excellent post by Lisa Charlotte Rost : *http://bit.ly/2gRGx1J*

(figures taken from her post)



Tableau

Important Details on Tableau Public

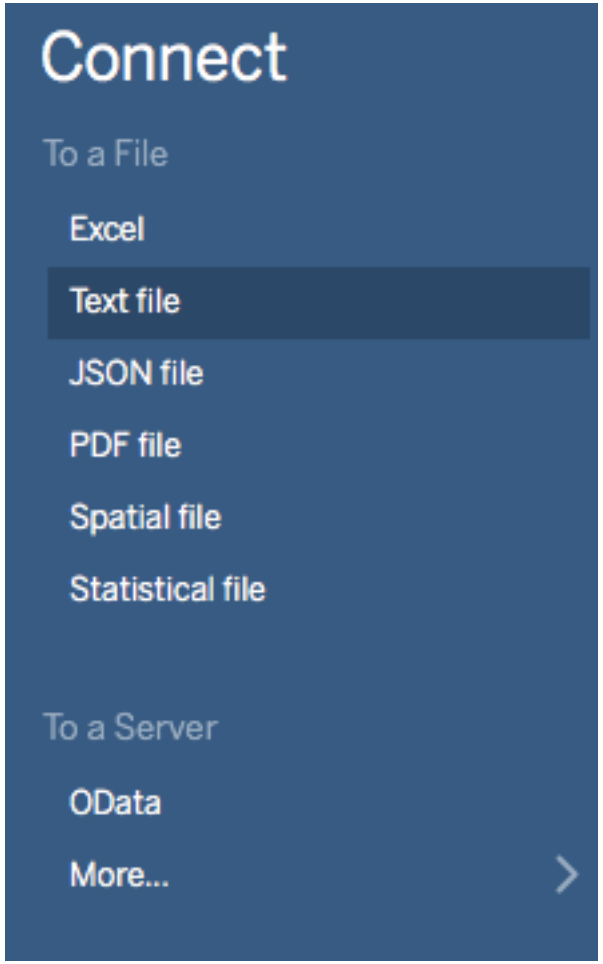
- **I am not a Tableau Expert**
  - There are plenty of great training videos & tutorials online
  - I mainly use R, and rarely need to use Tableau
  - I do, however, find Tableau pretty easy to use
- **I'll be demonstrating using Tableau Public**
  - Some features are different than the version of Tableau you pay for
- **YOU CANNOT PRIVATELY SAVE WORKBOOKS USING TABLEAU PUBLIC**
  - Please don't use Tableau *Public* for sensitive data, because anything you save will be publicly posted!
  - Tableau Desktop that you pay for *does* allow you to privately save workbooks

Today I'll demonstrate the following

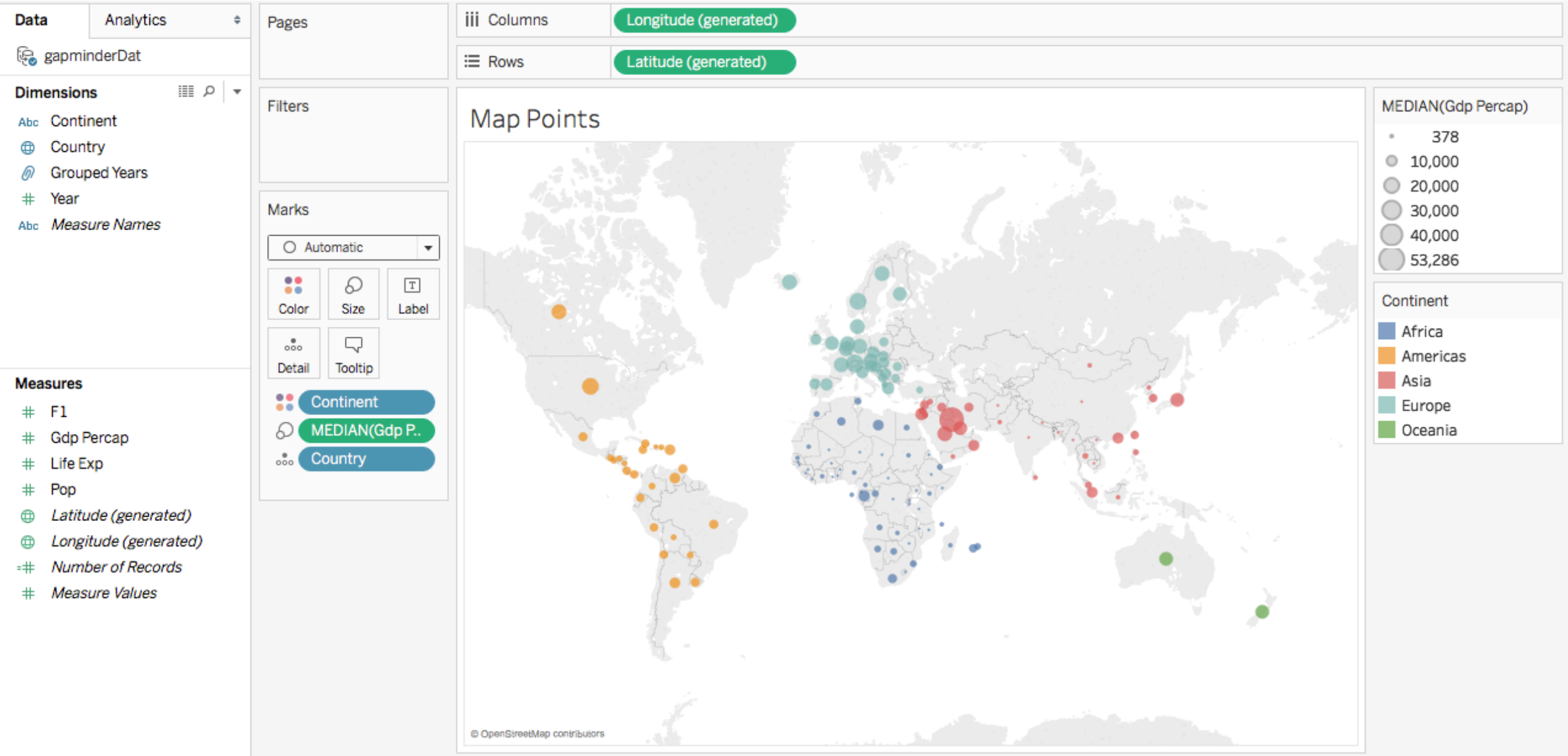
- Loading data
- Creating two types of maps
- Creating a scatter plot
- Exploring alternative visual designs with "Show Me"
- Putting it all together into a dashboard

I've put together some screen shots to help you reconstruct the Tableau analysis

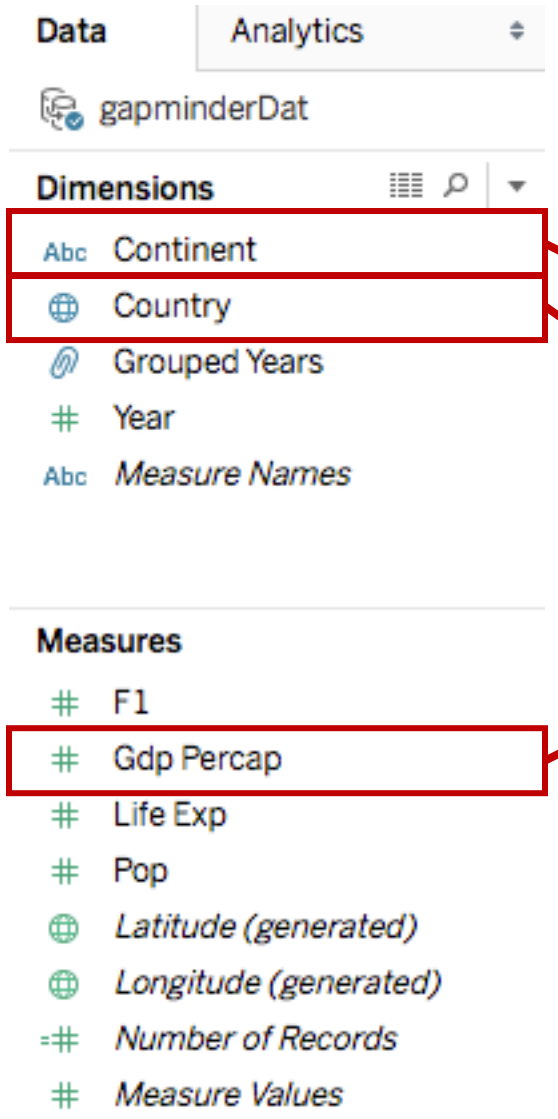
Step 1: Load Gapminder CSV into Tableau



Step 2: Create a map with median gdp per capita as dots

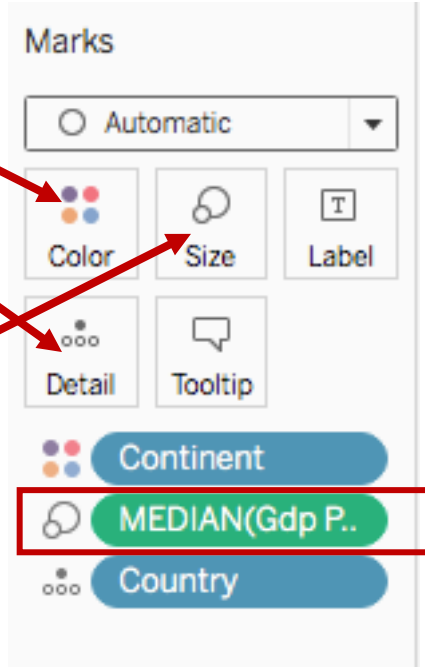


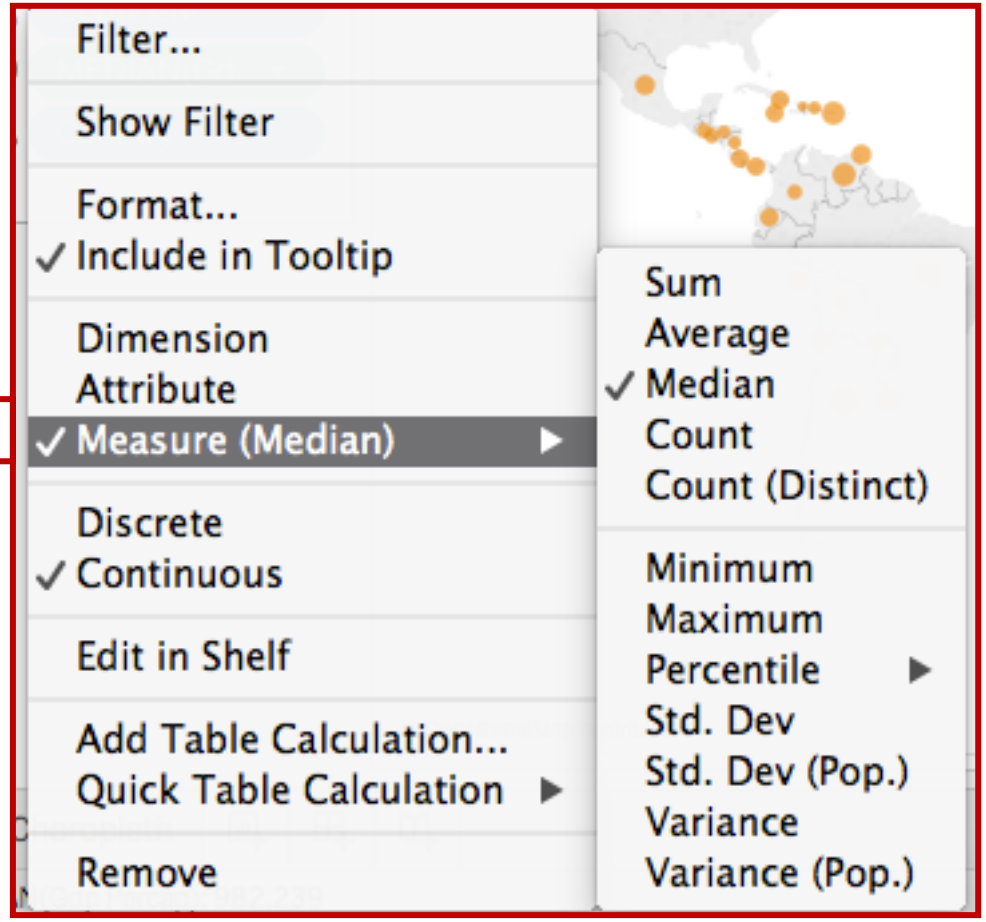
Step 2: Create a map with median gdp per capita as dots



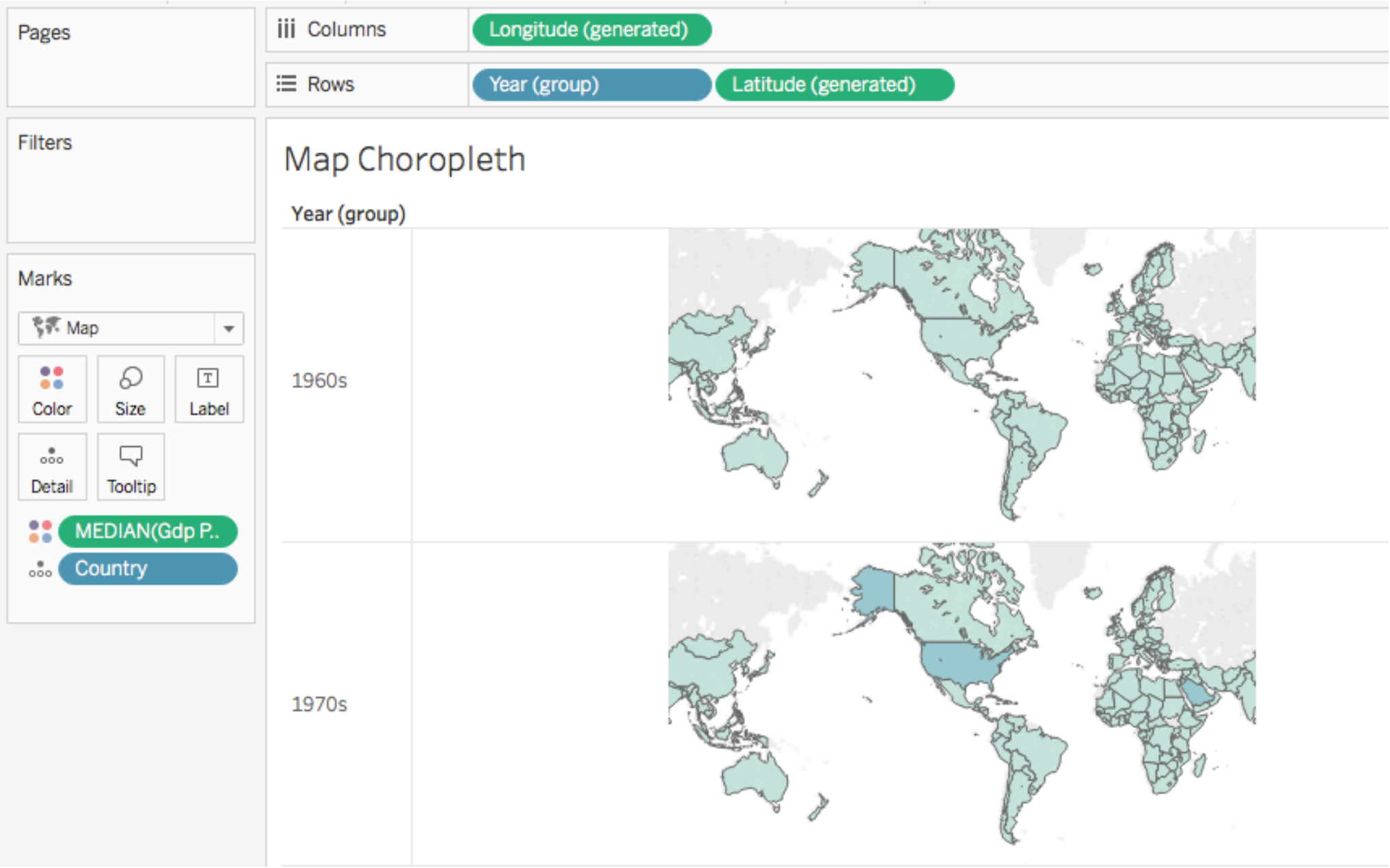
A: Drag Dimensions & measures to marks

B: Set gpdPerCaptial to median (sum is default)

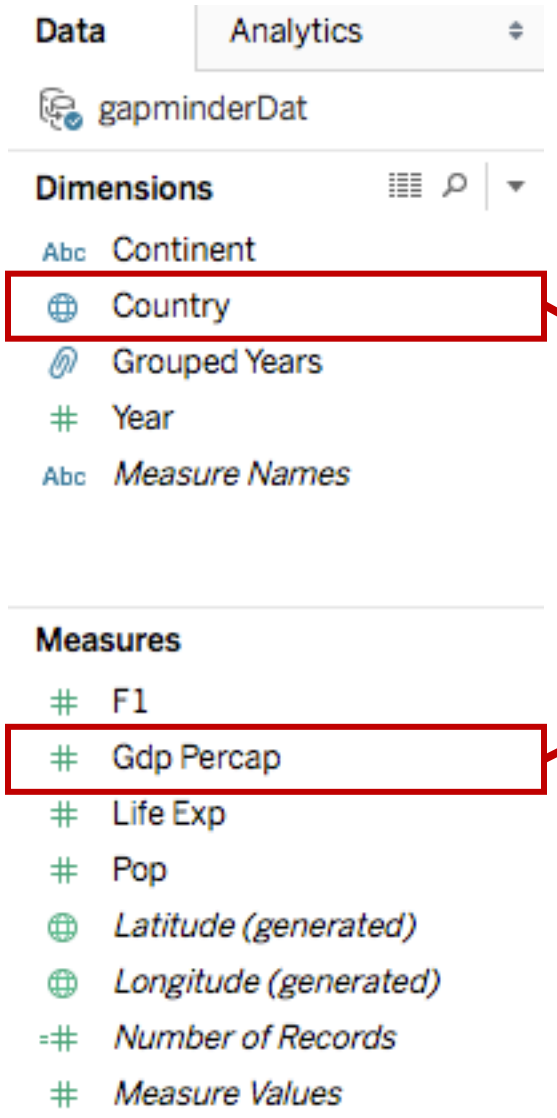




Step 3: Create a small multiples choropleth map



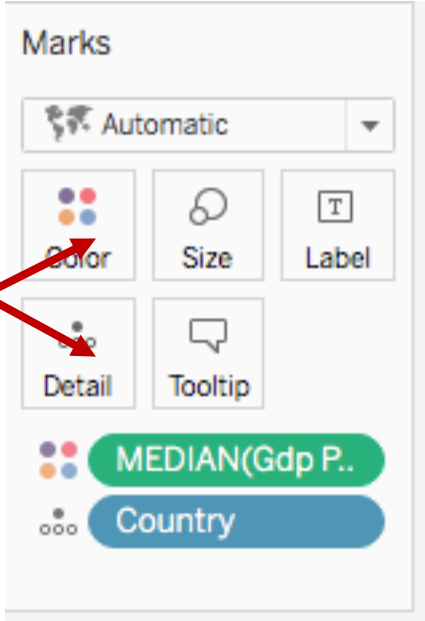
Step 3: Create a small multiples choropleth map



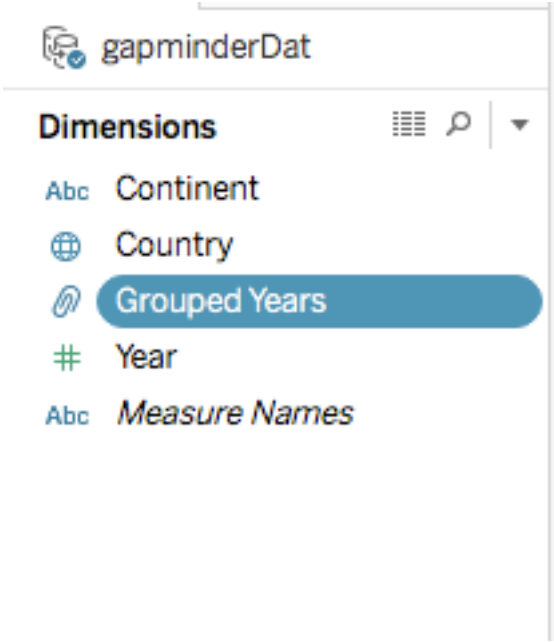
Drag to

Drag to

A: Drag Dimensions & measures to marks



B: *Create* a new group

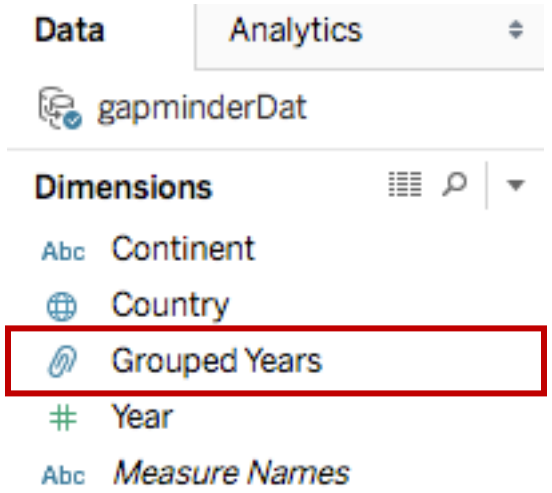


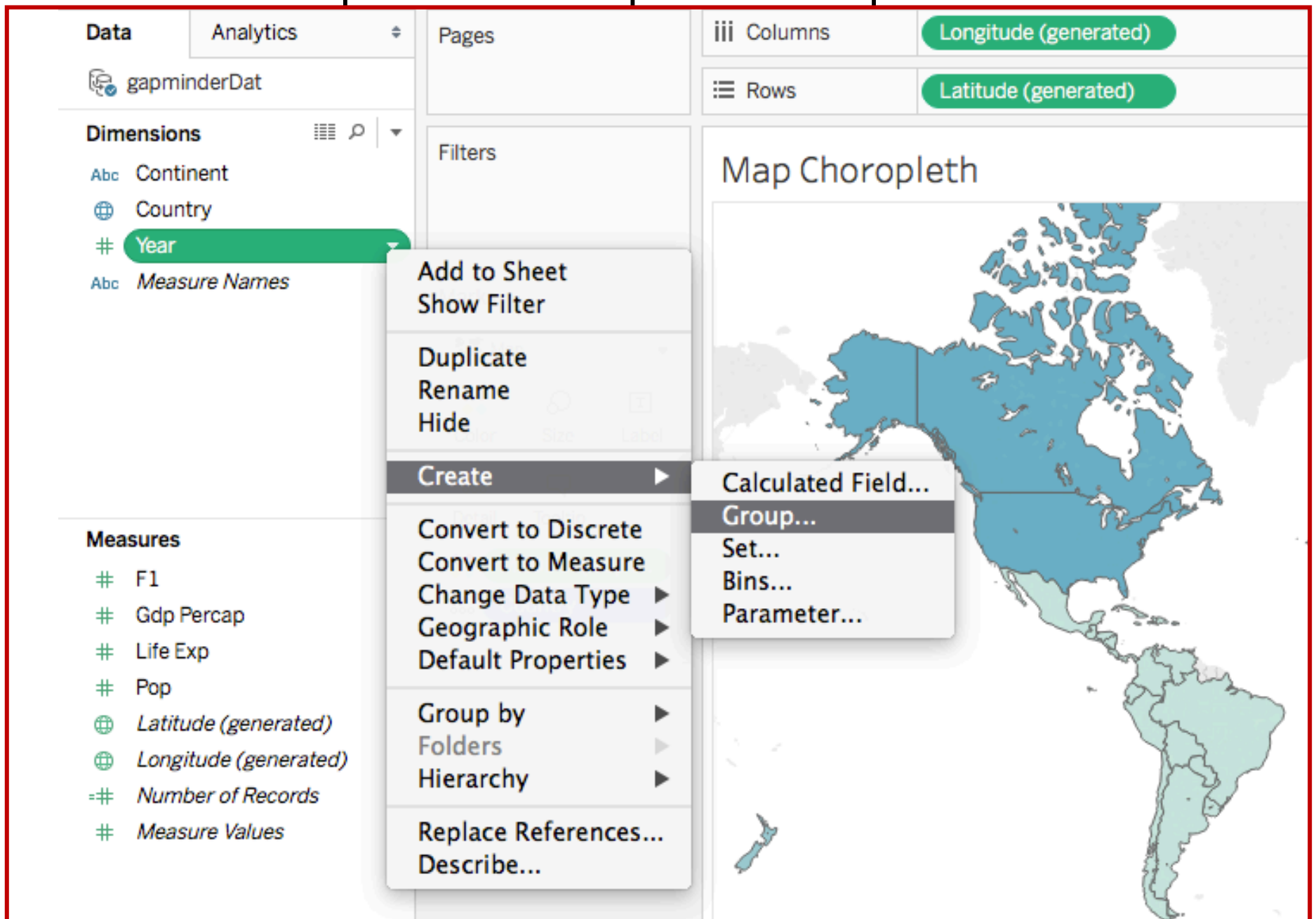
Same as before

New!

Step 3: Create a small multiples choropleth map

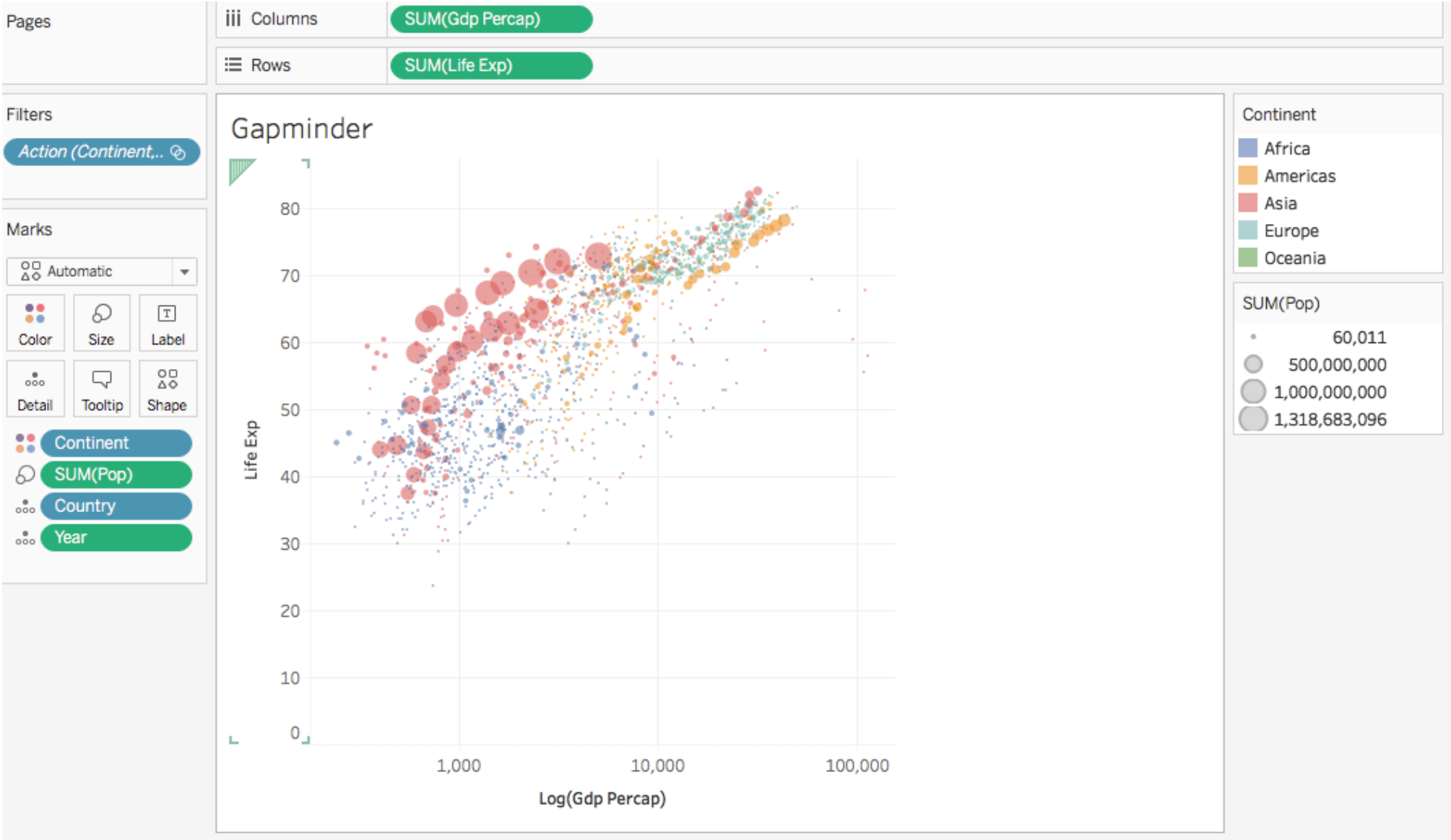
B: *Create* a new group





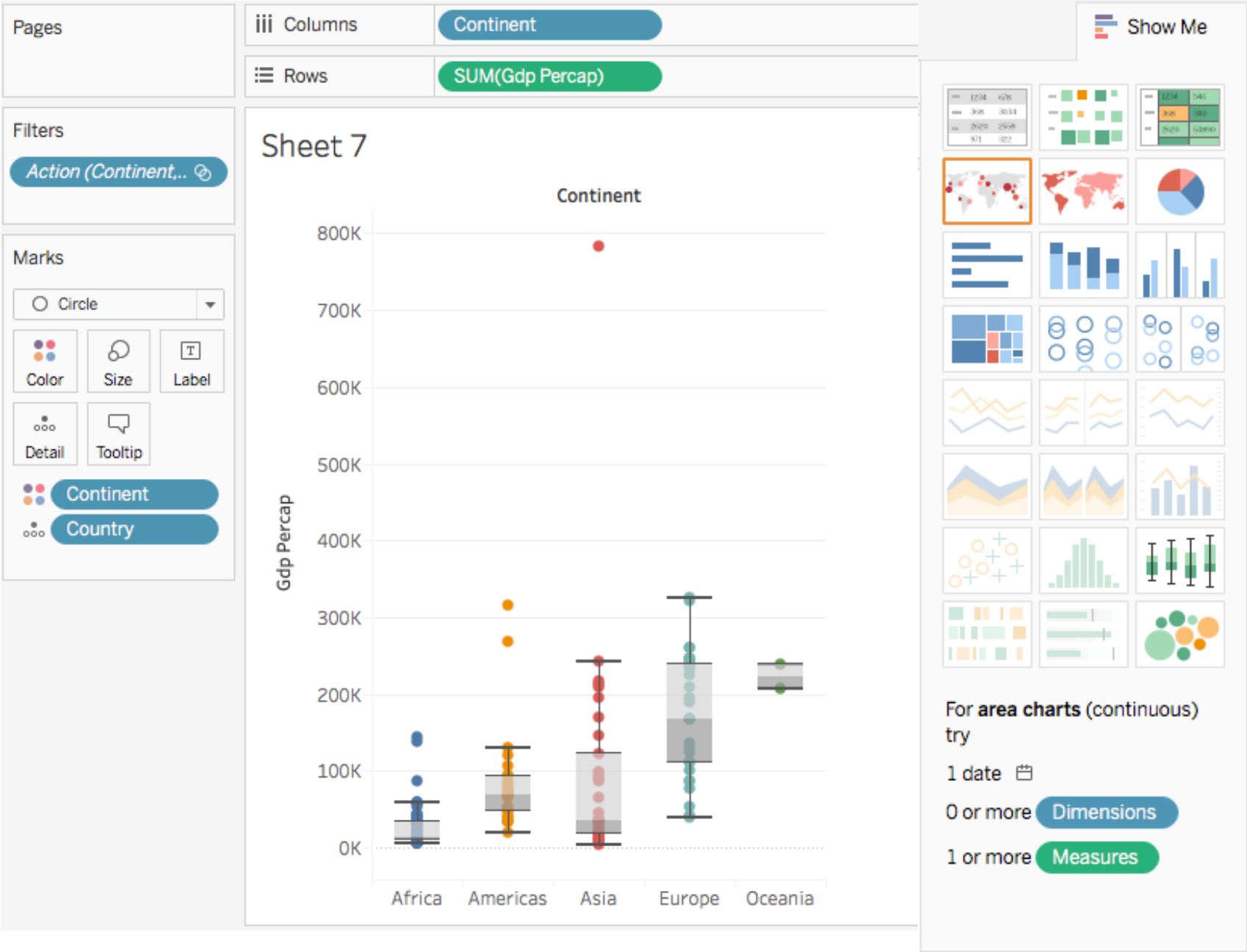
Step 4: Create a scatter plot

Try it on your own

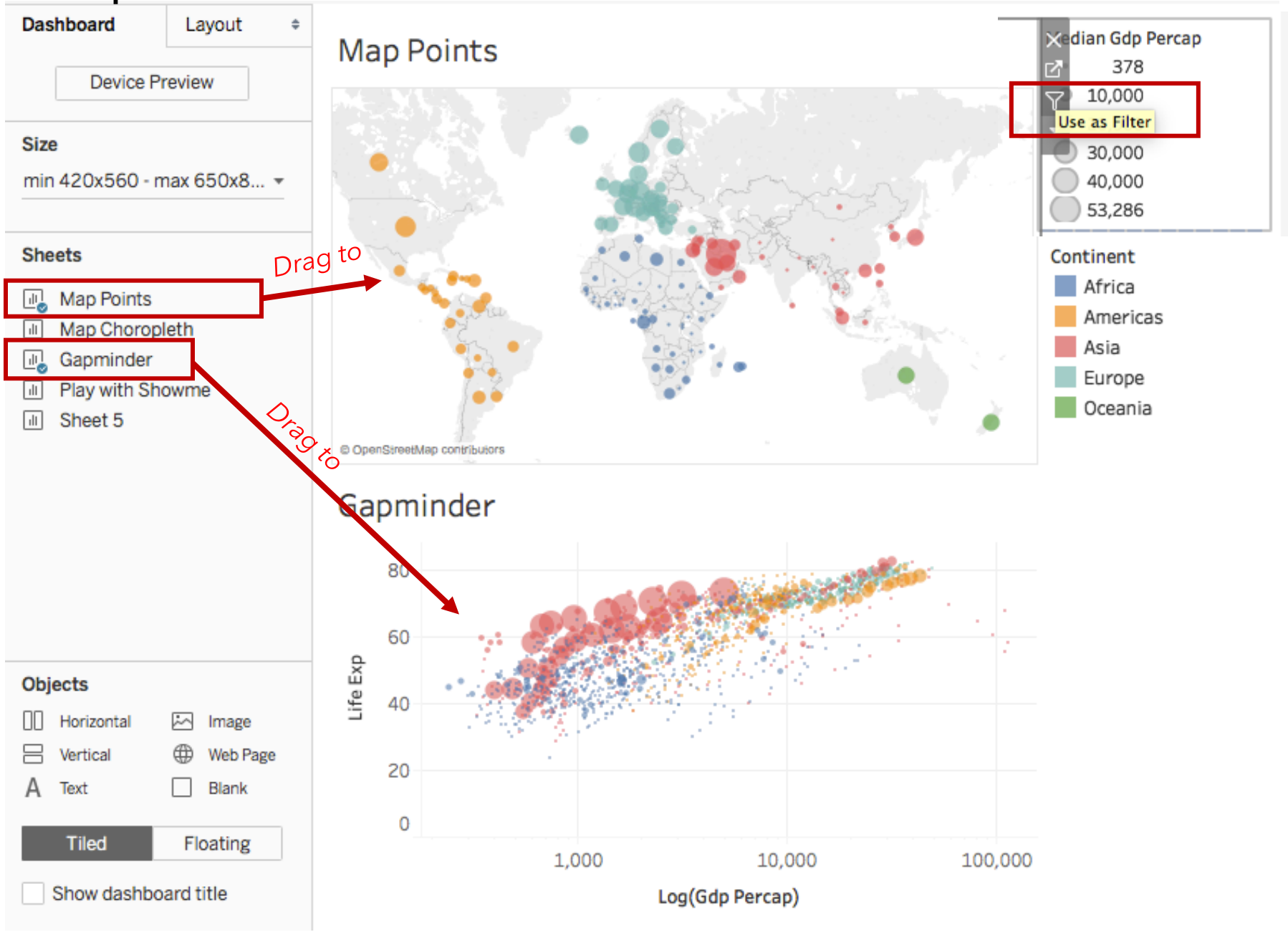


Step 5: Explore the ShowMe panel

Try it on your own



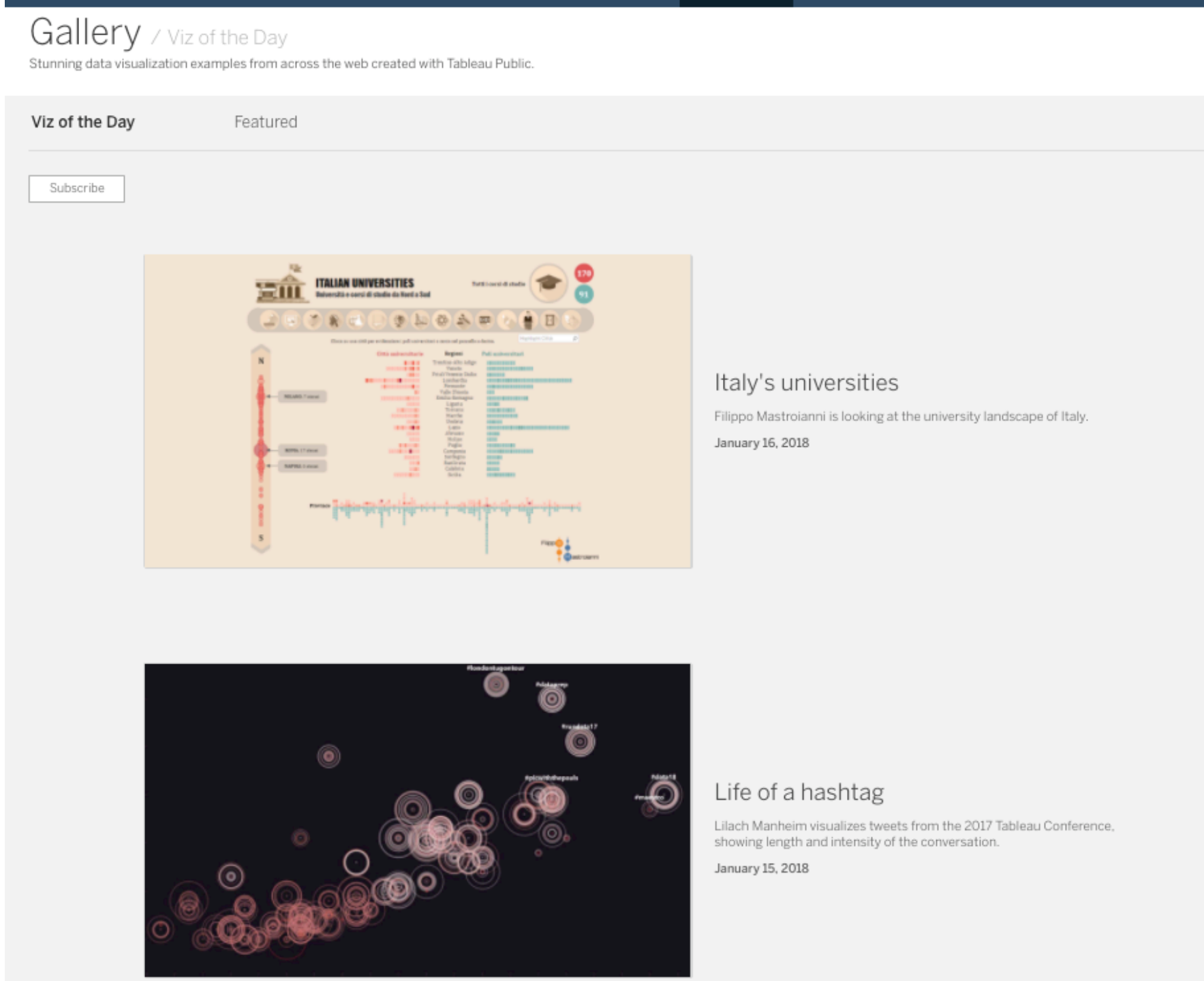
Step 6: Create a dashboard



Create a filter

See more online at the Tableau Public gallery!

https://public.tableau.com/en-us/s/gallery

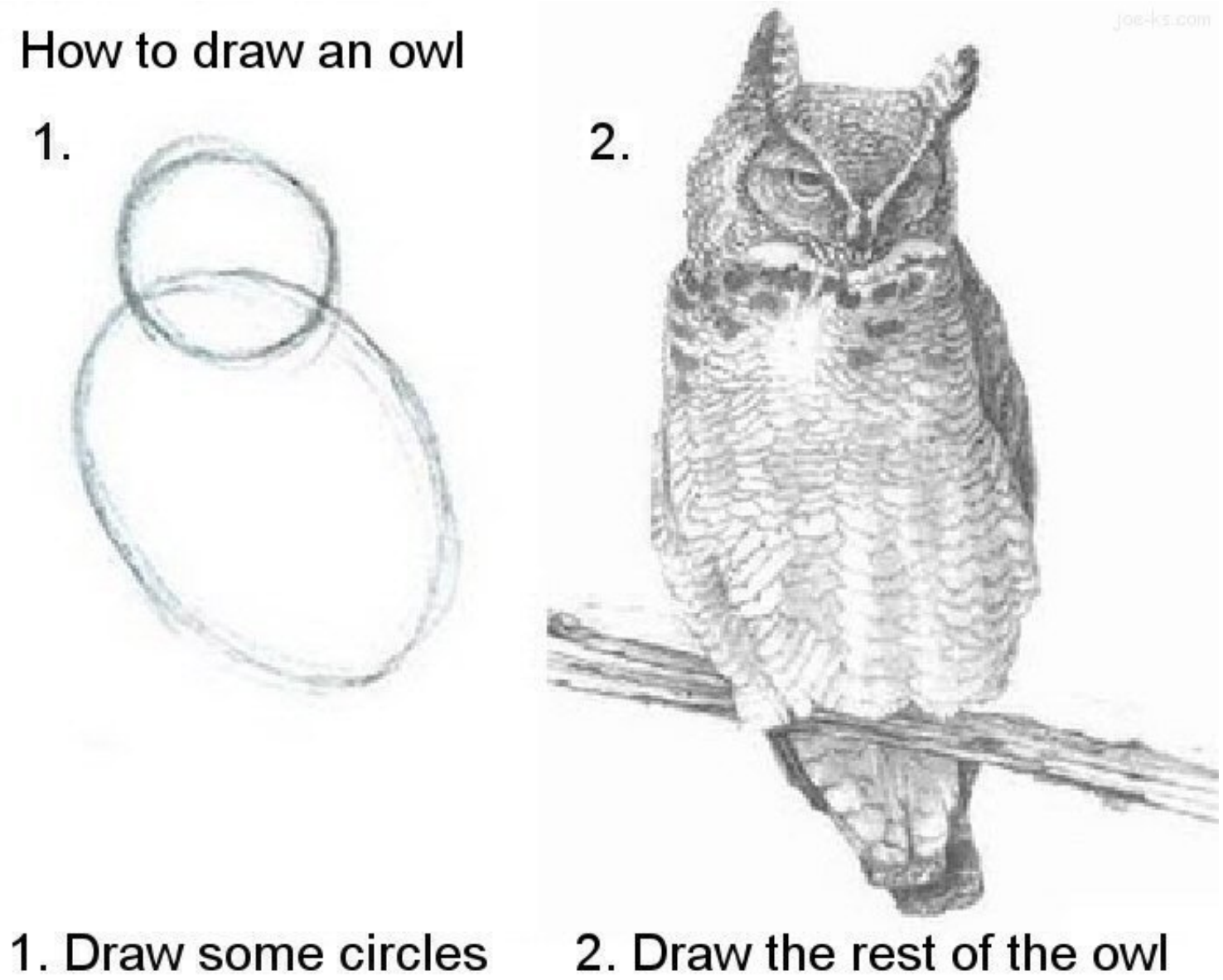


Shiny

Important Details on R and Shiny

- To introduce shiny, I am using selected slide content from several presentations developed for the "Shiny quick start guide" available here:
  - Part 1: bit.ly/shiny-quickstart-1
  - Part 2: bit.ly/shiny-quickstart-2
- These quick start slides are also presented as a video on the Shiny website
  - https://shiny.rstudio.com/tutorial/

The Challenges of Briefly Introducing Shiny



- Shiny is powerful and somewhat simpler than web languages - but it still takes some time to learn it
- I want to show you some basic info, and give you a sense of what Shiny can do - but there is a lot of in between I won't cover

What is Shiny?

HTML

Basic Structure

CSS

Presentation, Layouts, formatting

JS

Webpage / Application Behavior

+

Most glorious of programming languages

How Does Shiny Work?

Every Shiny app is maintained by a computer running R

Server runs R code

User Interface (UI)

bit.ly/shiny-quickstart-1

Basic Structure of a Shiny Application

```
library(shiny)
ui <- fluidPage()

server <- function(input, output) {}

shinyApp(ui = ui, server = server)
```

bit.ly/shiny-quickstart-1

Basic Structure of a Shiny Application

An entire application can be a single file, or broken down into a ui.R & server.R files

```r
library(shiny)

ui <- fluidPage(
  sliderInput(inputId = "num",
    label = "Choose a number",
    value = 25, min = 1, max = 100),
  plotOutput("hist")
)

server <- function(input, output) {
  output$hist <- renderPlot({
    hist(rnorm(input$num))
  })
}

shinyApp(ui = ui, server = server)
```

```r
# ui.R
library(shiny)
fluidPage(
  sliderInput(inputId = "num",
    label = "Choose a number",
    value = 25, min = 1, max = 100),
  plotOutput("hist")
)
```

```r
# server.R
library(shiny)
function(input, output) {
  output$hist <- renderPlot({
    hist(rnorm(input$num))
  })
}
```

You **must** use **app.r** (single file); or **ui.r** and **server.r** names, or your application won't run

The user interface (ui.r) houses inputs & outputs

Example of inputs

Buttons

Action

Submit

`actionButton()`
`submitButton()`

Single checkbox

☑ Choice A

`checkboxInput()`

Checkbox group

☑ Choice 1
☐ Choice 2
☐ Choice 3

`checkboxGroupInput()`

Date input

2014-01-01

`dateInput()`

Date range

2014-01-24 to 2014-01-24

`dateRangeInput()`

File input

Choose File No file chosen

`fileInput()`

Numeric input

1

`numericInput()`

Password Input

••••••••••

`passwordInput()`

Radio buttons

◉ Choice 1
○ Choice 2
○ Choice 3

`radioButtons()`

Select box

Choice 1

`selectInput()`

Sliders

0 50 100

0 25 75 100

`sliderInput()`

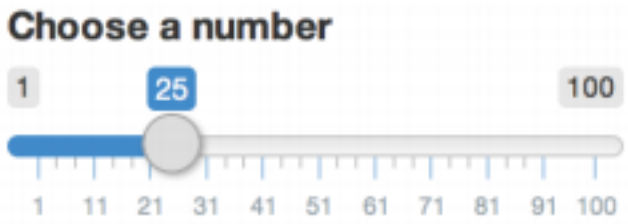
Text input

Enter text...

`textInput()`

The user interface (ui.r) houses inputs & outputs

Example of inputs




```
sliderInput(inputId = "num", label = "Choose a number", …)
```

input name (for internal use)

Notice: Id not ID

label to display

input specific arguments

Changing the slider number (i.e. dragging the circle left to right) causes R to automatically initiate an action to update all outputs that use the slider number.

bit.ly/shiny-quickstart-1

The user interface (ui.r) houses inputs & outputs

Example of outputs

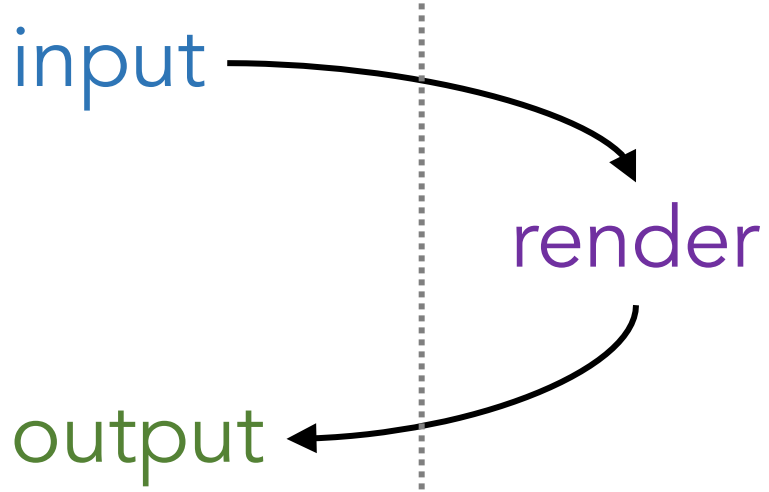
Function	Inserts
`dataTableOutput()`	an interactive table
`htmlOutput()`	raw HTML
`imageOutput()`	image
`plotOutput()`	plot
`tableOutput()`	table
`textOutput()`	text
`uiOutput()`	a Shiny UI element
`verbatimTextOutput()`	text

Function = command telling R & shiny to do some particular task

server.r stiches inputs & outputs together

ui.r
Shiny widgets
Interface elements

server.r
ggplot code
dataset manipulation

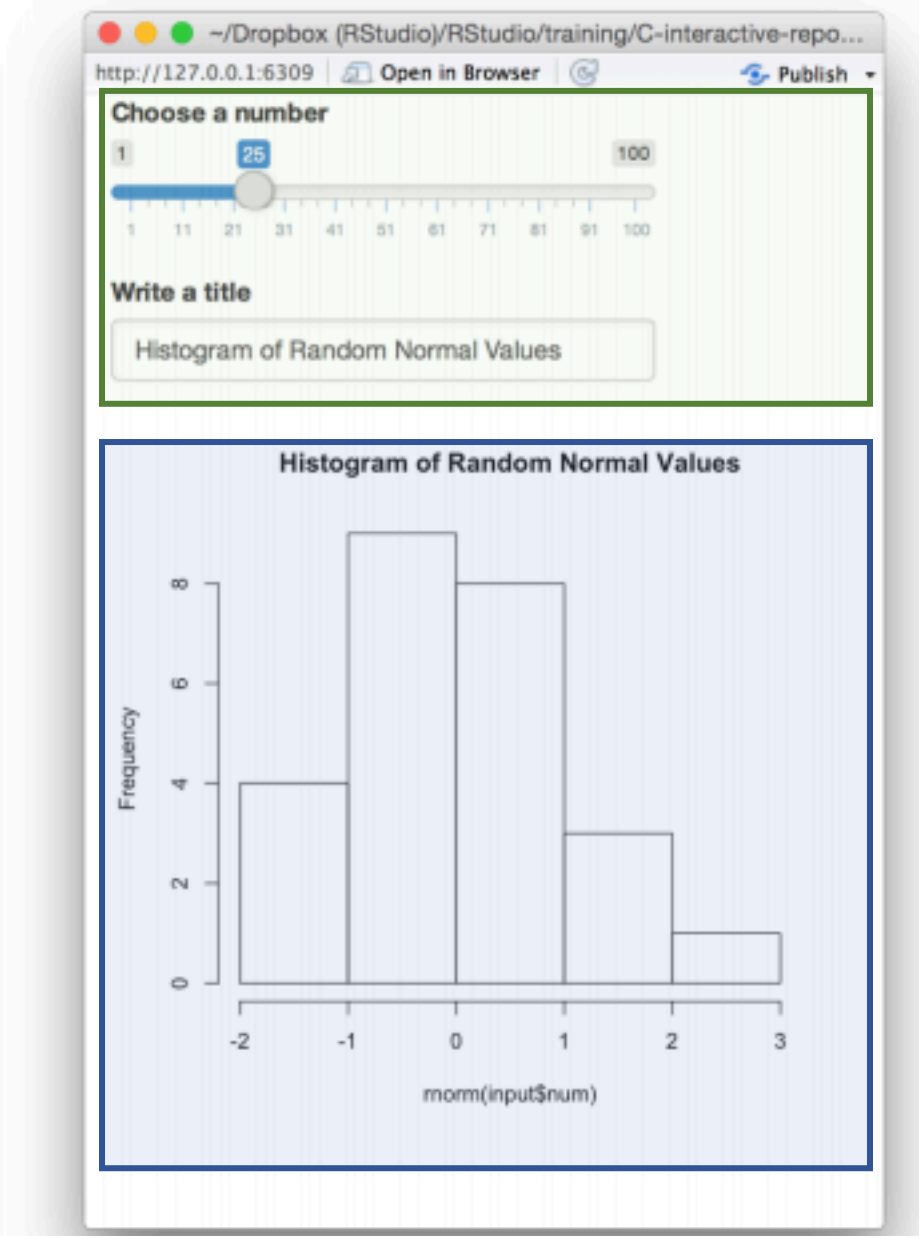


Output & render functions are complimentary

output function	render function
plotOutput( )	renderPlot( )
tableOutput( )	renderTable( )

A simple worked example

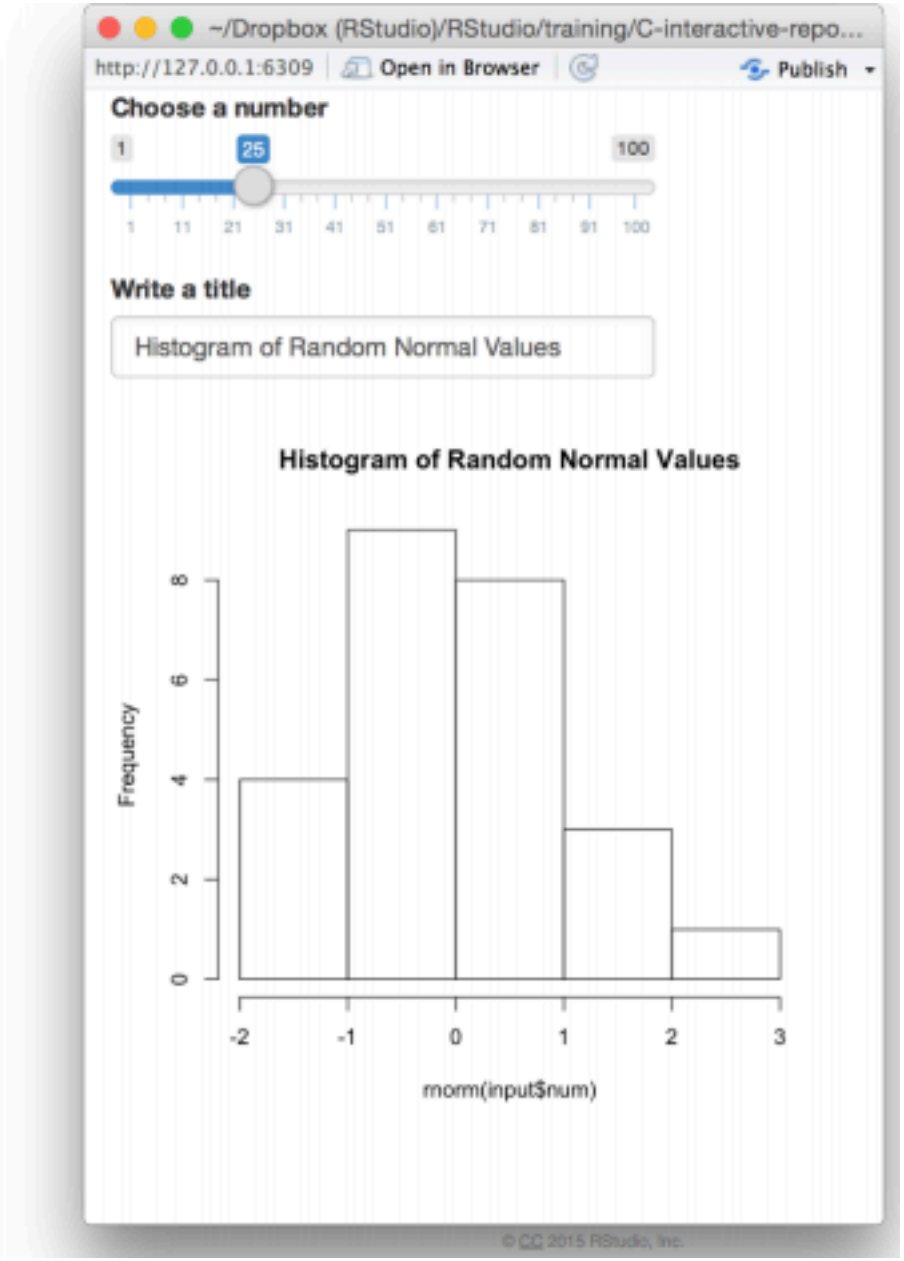
```
# 01-two-inputs

library(shiny)

ui <- fluidPage(
  sliderInput(inputId = "num",
    label = "Choose a number",
    value = 25, min = 1, max = 100),
  textInput(inputId = "title",
    label = "Write a title",
    value = "Histogram of Random Normal Values"),
  plotOutput("hist")
)

server <- function(input, output) {
  output$hist <- renderPlot({
    hist(rnorm(input$num), main = input$title)
  })
}

shinyApp(ui = ui, server = server)
```



bit.ly/shiny-quickstart-1

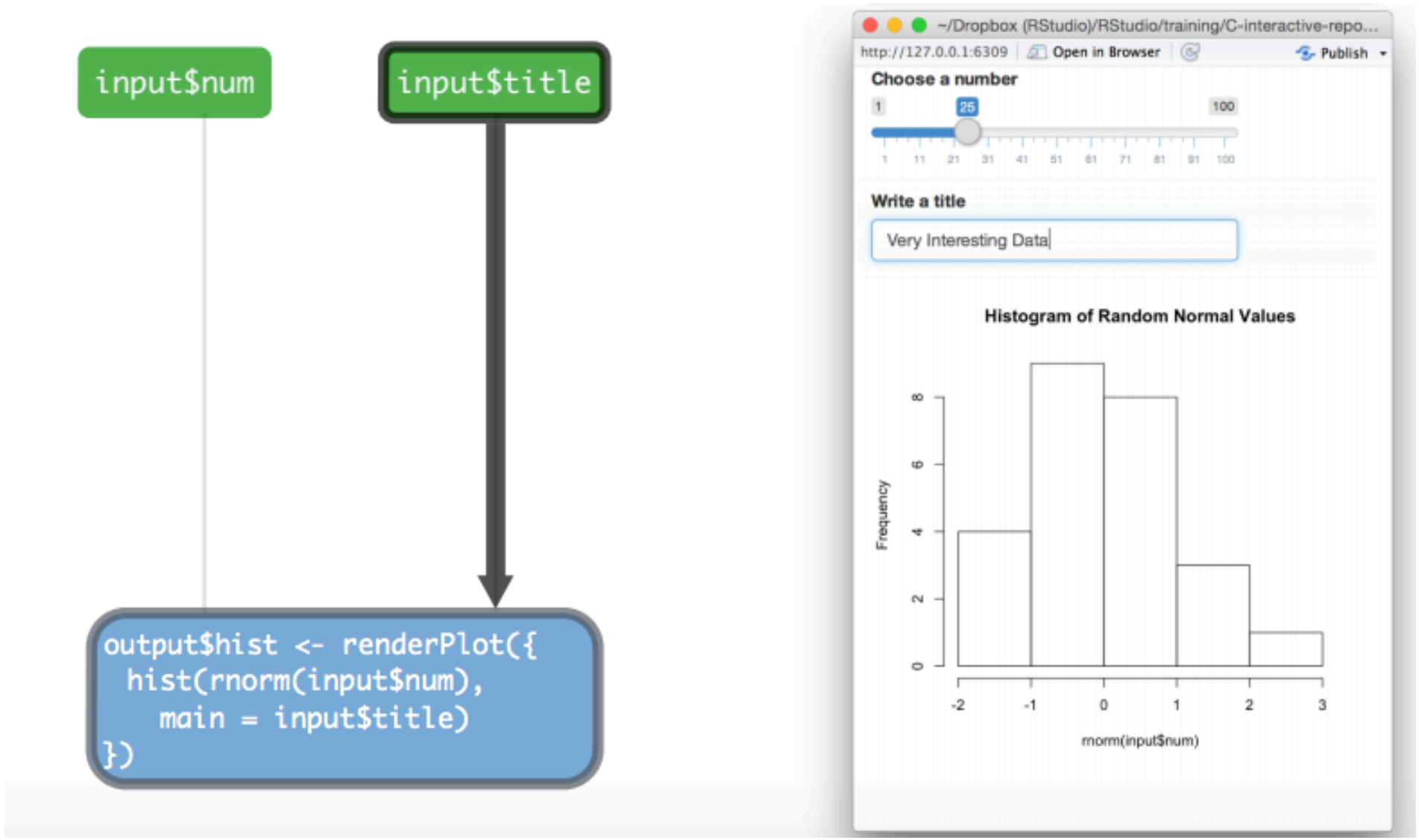
Reactivity in Shiny – a Very Brief Introduction

```r
# 01-two-inputs

library(shiny)

ui <- fluidPage(
  sliderInput(inputId = "num",
    label = "Choose a number",
    value = 25, min = 1, max = 100),
  textInput(inputId = "title",
    label = "Write a title",
    value = "Histogram of Random Normal Values"),
  plotOutput("hist")
)

server <- function(input, output) {
  output$hist <- renderPlot({
    hist(rnorm(input$num), main = input$title)
  })
}

shinyApp(ui = ui, server = server)
```



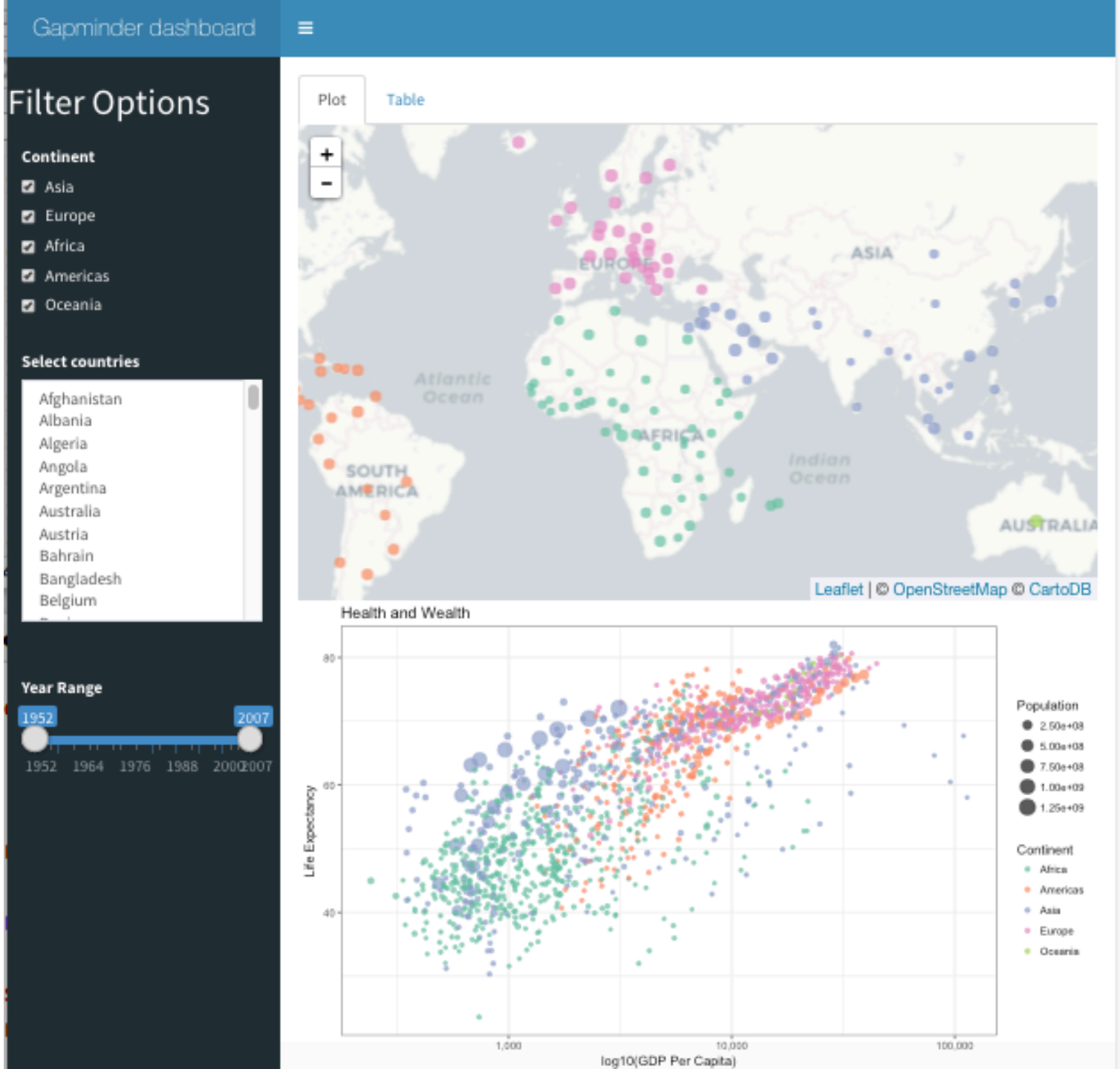
bit.ly/shiny-quickstart-1

Reactivity in Shiny – a Very Brief Introduction



bit.ly/shiny-quickstart-1

Let’s go through a more complex example



See more online at the Shiny Gallery!

https://www.rstudio.com/products/shiny/shiny-user-showcase/

RStudio

Products Resources Pricing About Us Blog

Shiny Apps for the Enterprise

Shiny Dashboard Demo

A dashboard built with Shiny.

Location tracker

Track locations over time with streaming data.

Download monitor

Streaming download rates visualized as a bubble chart.

Supply and Demand

Forecast demand to plan resource allocation.

Industry Specific Shiny Apps

Economic Dashboard

Economic forecasting with macroeconomic indicators.

ER Optimization

An app that models patient flow.

CDC Disease Monitor

Alert thresholds and automatic weekly updates.

Ebola Model

An epidemiological simulation.